

# Connor W. Magoon

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| Education    | <p><b>The University of North Carolina at Chapel Hill (UNC)</b></p> <p>PhD Applied Mathematics 2023–2028</p> <p>Thesis advisor: <a href="#">Pedro Sáenz</a>, Secondary advisor: <a href="#">Shahar Kovalsky</a></p> <p>BS Physics and Applied Mathematics 2019–2023</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Positions    | <p><b>Simons Foundation</b> Jun–Aug 2026</p> <p>Summer Pre-Doctoral Researcher</p> <p>Flatiron Institute / Center for Computational Biology / Biophysical Modeling Group</p> <p>Mentors: Scott Weady, David Stein</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Publications | <p><b>Differentiation Through Black-Box Quadratic Programming Solvers</b></p> <p><a href="#">C. W. Magoon*</a>, <a href="#">F. Yang*</a>, <a href="#">N. Aigerman</a>, <a href="#">S. Z. Kovalsky</a></p> <p><i>NeurIPS</i>, 2025   <a href="#">Code</a>   <a href="#">Poster</a></p> <p><b>The way bubbles gallop</b></p> <p><a href="#">J. H. Guan</a>, <a href="#">S. I. Tamim</a>, <a href="#">C. W. Magoon</a>, <a href="#">H. A. Stone</a>, and <a href="#">P. J. Sáenz</a></p> <p><i>Physical Review Fluids</i>, 2025   <a href="#">Video</a></p> <p><b>Galloping Bubbles</b></p> <p><a href="#">J. H. Guan*</a>, <a href="#">S. I. Tamim*</a>, <a href="#">C. W. Magoon*</a>, <a href="#">H. A. Stone</a>, and <a href="#">P. J. Sáenz</a></p> <p><i>Nature Communications</i>, 2025</p> <p><b>Traveling Faraday waves</b></p> <p><a href="#">J. H. Guan</a>, <a href="#">C. W. Magoon</a>, <a href="#">M. Durey</a>, <a href="#">R. Camassa</a>, and <a href="#">P. J. Sáenz</a></p> <p><i>Physical Review Fluids</i>, 2023   <a href="#">Video</a></p> <p>*Equal contribution</p> |
| Submitted    | <p><b>Spontaneous galloping of a vibrating bubble along a solid boundary. Part 1: Experiments</b></p> <p><a href="#">J. H. Guan</a>, <a href="#">X. Liu</a>, <a href="#">S. I. Tamim</a>, <a href="#">C. W. Magoon</a>, and <a href="#">P. J. Sáenz</a></p> <p><b>Spontaneous galloping of a vibrating bubble along a solid boundary. Part 2: Simulations</b></p> <p><a href="#">C. W. Magoon</a>, <a href="#">S. I. Tamim</a>, and <a href="#">P. J. Sáenz</a></p> <p><b>Spontaneous galloping of a vibrating bubble along a solid boundary. Part 3: Theory</b></p> <p><a href="#">S. I. Tamim</a>, <a href="#">C. W. Magoon</a>, and <a href="#">P. J. Sáenz</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Awards       | <p><i>Travel Grants</i></p> <p><b>NeurIPS Scholar Award</b> 2025</p> <p><i>Conference Awards</i></p> <p><b>Gallery of Fluid Motion Award</b> 2024</p> <p><a href="#">J. H. Guan</a>, <a href="#">S. I. Tamim</a>, <a href="#">C. W. Magoon</a>, <a href="#">H. A. Stone</a>, and <a href="#">P. J. Sáenz</a>, “Galloping Bubbles”</p> <p><i>American Physical Society / Division of Fluid Dynamics</i></p> <p><b>Milton van Dyke Award</b> 2022</p> <p><a href="#">J. H. Guan</a>, <a href="#">C. W. Magoon</a>, <a href="#">M. Durey</a>, <a href="#">R. Camassa</a>, and <a href="#">P. J. Sáenz</a>, “Run, Faraday, Run”</p> <p><i>American Physical Society / Division of Fluid Dynamics</i></p> <p><i>Honors</i></p> <p><b>Doctoral Merit Fellowship, UNC</b> 2023–2024</p> <p><b>Archibald Henderson Mathematics Medal, UNC</b> 2023</p>                                                                                                                                                                                                                                              |

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| Talks    | <b>Collective Dynamics of Galloping Bubbles</b>                                                 | 2025            |
|          | C. W. Magoon, X. Liu, S. I. Tamim, H. A. Stone and P. J. Sáenz                                  |                 |
|          | APS Division of Fluid Dynamics Meeting                                                          |                 |
|          | <b>Collective Dynamics of Galloping Bubbles</b>                                                 | 2024            |
|          | C. W. Magoon, X. Liu, J. H. Guan, S. I. Tamim, and P. J. Sáenz                                  |                 |
|          | APS Division of Fluid Dynamics Meeting                                                          |                 |
|          | <b>Direct Numerical Simulations of Vibrating Bubbles Self-Propelling Along Solid Boundaries</b> | 2023            |
|          | C. W. Magoon, J. H. Guan, S. I. Tamim, and P. J. Sáenz                                          |                 |
|          | APS Division of Fluid Dynamics Meeting                                                          |                 |
| Teaching | <i>Instructor</i>                                                                               |                 |
|          | <b>Differential Equations Lab</b> (UNC MATH 383L)                                               | Fall 2025       |
|          | <i>Teaching Assistant</i>                                                                       |                 |
|          | <b>Multivariable Calculus</b> (UNC MATH 233)                                                    | Fall 2024, 2025 |
|          | <b>General Physics</b> (UNC PHYS 115)                                                           | Summer 2021     |
|          | <b>Introductory Electromagnetism</b> (UNC PHYS 119)                                             | Fall 2020       |
|          | <i>Grader</i>                                                                                   |                 |
|          | <b>Fluid Mechanics</b> (UNC MATH 563)                                                           | Spring 2025     |
|          | <b>Differential Equations</b> (UNC MATH 524)                                                    | Spring 2024     |